MH1100 Calculus I

Handout 10 - Application of Differentiation II

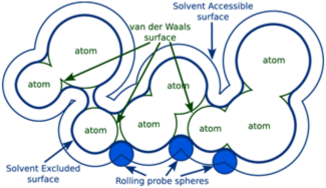
Synopsis

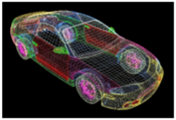
Application of Differentiation

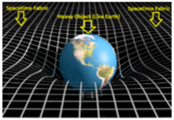
- Derivatives and Shape of a Graph
- Limits at Infinity
- Horizontal Asymptotes
- Slant Asymptotes
- Guidelines for Sketching a Curve

Further Reading:

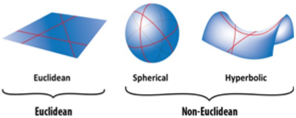
Differential geometry;
Riemannian geometry; Manifold;
Curvatures; Shape analysis;
Geometric modeling; computer-aided geometric design...

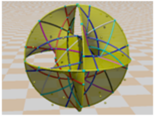


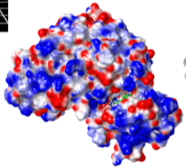


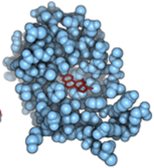


Geometric modelling of protein surface









Derivatives and Shape of a Graph; Limits at Infinity; Horizontal Asymptotes; Slant Asymptotes; Guidelines for Sketching a Curve

What Does f' Say About f

Derivative of $f(x)$ can tell us where a function is increasing or decreasing!!

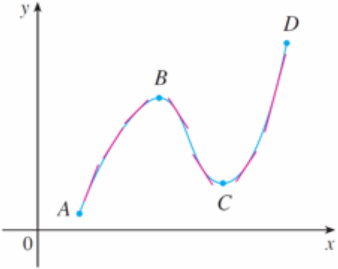


Figure 1

Between A and B and between C and D, tangent lines have positive slopes, so $f'(x) > 0$. At the same time, $f(x)$ is increasing.
Between B and C, tangent lines have negative slope, i.e., $f'(x) < 0$. At the same time, $f(x)$ is decreasing.

To prove that this is always the case, we use the Mean Value Theorem.

Increasing/Decreasing (I/D) Test

(a) If $f'(x) > 0$ on an interval, then $f(x)$ is increasing on that interval

(b) If $f'(x) < 0$ on an interval, then $f(x)$ is decreasing on that interval

Prove: (a) Let x_1 and x_2 be any two numbers in the interval with $x_1 < x_2$. We need to show if $f'(x) > 0$, the $f(x)$ is increasing, i.e., $f(x_1) < f(x_2)$. Since $f'(x)$ exists on $[x_1, x_2]$, $f(x)$ is continuous on $[x_1, x_2]$ and differentiable on (x_1, x_2), using Mean Value Theorem, we have

convert property of derivative to property of function

$\exists\, c \in (x_1, x_2)$

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$$

with $x_1 < c < x_2$. As $f'(c) > 0$ and $x_2 - x_1 > 0$, we have $f(x_1) < f(x_2)$. The function is increasing on the interval.

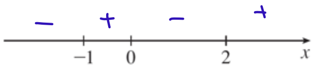
What Does f' Say About f

Find where the function $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ is increasing and where it is decreasing.

Solution:
We start by differentiating $f(x)$,

$$f'(x) = 12x^3 - 12x^2 - 24x = 12x(x-2)(x+1).$$

To use the Increasing/Decreasing Test, we have to know where $f'(x) > 0$ and where $f'(x) < 0$. We first find where $f'(x) = 0$. They are $x = 0$, 2, and -1. These are the critical numbers of $f(x)$, and they divide the domain into four intervals. Within each interval, $f'(x)$ is always positive or always negative!!



What Does f' Say About f

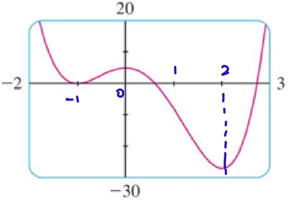
For each interval, we determine the signs of the three factors of $f'(x)$, namely, $12x$, $x - 2$, and $x + 1$. (A plus sign indicates that the given expression is positive and a minus sign indicates that it is negative.)

Interval	$12x$	$x - 2$	$x + 1$	$f'(x)$	f
$x < -1$	$-$	$-$	$-$	$-$	decreasing on $(-\infty, -1)$
$-1 < x < 0$	$-$	$-$	$+$	$+$	increasing on $(-1, 0)$
$0 < x < 2$	$+$	$-$	$+$	$-$	decreasing on $(0, 2)$
$x > 2$	$+$	$+$	$+$	$+$	increasing on $(2, \infty)$

The last column of the chart gives the conclusion based on the I/D Test. For instance, $f'(x) < 0$ for $0 < x < 2$, so $f(x)$ is decreasing on $(0, 2)$ (or $[0, 2]$).

What Does f' Say About f

The graph of f confirms our analysis.



Interval	$12x$	$x-2$	$x+1$	$f'(x)$	f
$x<-1$	−	−	−	−	decreasing on $(-\infty,-1)$
$-1<x<0$	−	−	+	+	increasing on $(-1,0)$
$0<x<2$	+	−	+	−	decreasing on $(0,2)$
$x>2$	+	+	+	+	increasing on $(2,\infty)$

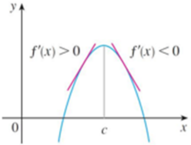
Here $f(0)=5$ is a local maximum value of $f(x)$, because $f(x)$ increases on $(-1,0)$ and decreases on $(0,2)$.
In other words, the sign of $f'(x)$ changes from positive to negative at 0.
This observation is the basis of the following test.

The First Derivative Test Suppose c is a critical number of a continuous function f.

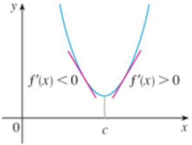
(a) If f' changes from positive to negative at c, then f has a local maximum at c.
$x < c$, $f'(x) > 0 \Rightarrow f(x)$ increasing
$x > c$, $f'(x) < 0 \Rightarrow f(x)$ decreasing

(b) If f' changes from negative to positive at c, then f has a local minimum at c.
$x < c$, $f'(x) < 0$
$x > c$, $f'(x) > 0$

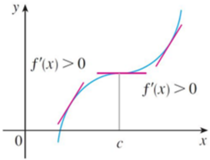
(c) If f' is positive to the right and left of c, or negative to the left and right of c, then f has no local maximum or minimum at c.



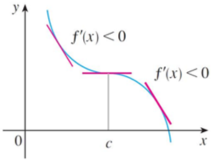
Local maximum
Figure 3(a)



Local minimum
Figure 3(b)



No maximum or minimum
Figure 3(c)



No maximum or minimum
Figure 3(d)

The First Derivative Test

Example: Find the local maximum and minimum values of the function

$$g(x) = x + 2\sin x, \ 0 \leq x \leq 2\pi$$

take note of domain!

Solution:

We start by finding the critical numbers. The derivative is:

$$g'(x) = 1 + 2\cos x$$

$x = \frac{2\pi}{3} + 2n\pi$ or $x = \frac{4\pi}{3} + 2n\pi$

so $g'(x) = 0$ when $\cos x = -\frac{1}{2}$. The solutions are $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$ $(0 \leq x \leq 2\pi)$. Because g is differentiable everywhere, the only critical numbers are $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$. We split the domain into intervals according to the critical numbers. Within each interval, $g'(x)$ is either always positive or always negative and so we analyze g in the following table.

Interval	$g'(x) = 1 + 2\cos x$	g
$0 < x < 2\pi/3$	$+$	increasing on $(0, 2\pi/3)$
$2\pi/3 < x < 4\pi/3$	$-$	decreasing on $(2\pi/3, 4\pi/3)$
$4\pi/3 < x < 2\pi$	$+$	increasing on $(4\pi/3, 2\pi)$

The First Derivative Test

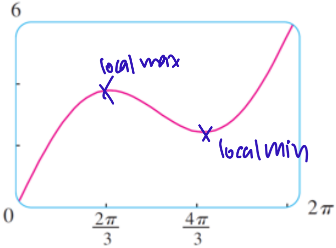
Because $g'(x)$ changes from positive to negative at $\frac{2\pi}{3}$, the First Derivative Test tells us that there is a local maximum at $\frac{2\pi}{3}$ and the local maximum value is

$$g(2\pi/3) = \frac{2\pi}{3} + 2\sin\frac{2\pi}{3} \approx 3.83$$

Likewise $g'(x)$ changes from negative to positive at $\frac{4\pi}{3}$, so

$$g(4\pi/3) = \frac{4\pi}{3} + 4\sin\frac{2\pi}{3} \approx 2.46$$

is a local minimum value.
The graph of $g(x)$ below supports our conclusion.

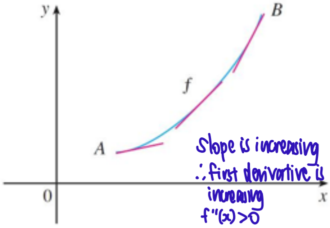


What Does f'' Say About f ?

In the Figure below, graphs of two increasing functions on (a, b). Both graphs join point A to point B but they look different because they bend in different directions. We check the tangents to these curves.

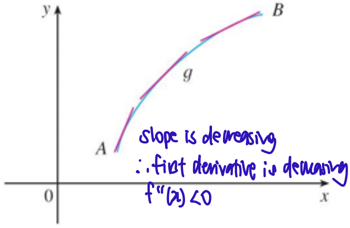
In Figure 6(a), the curve lies above the tangents and f is called concave upward on (a, b).

In Figure 6(b) the curve lies below the tangents and g is called concave downward on (a, b).



Concave upward

Figure 6(a)



Concave downward

Figure 6(b)

What Does f'' Say About f ?

Definition If the graph of f lies above all of its tangents on an interval I, then it is called **concave upward** in I. If the graph of f lies below all of its tangents on an interval I, then it is called **concave downward** in I.

Figure 7 shows the graph of a function that is concave upward (abbreviated CU) on the intervals (b, c), (d, e), and (e, p) and concave downward (CD) on the intervals (a, b), (c, d), and (p, q).

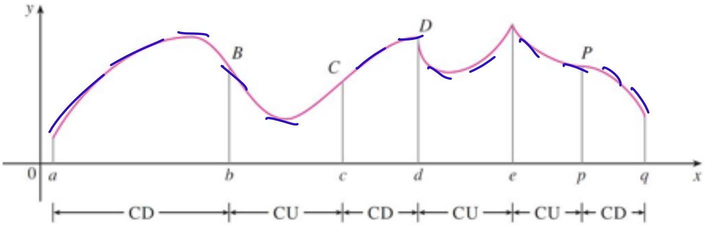
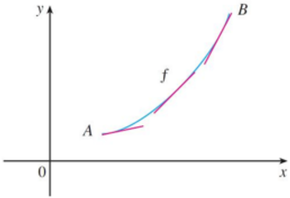


Figure 7

What Does f'' Say About f ?

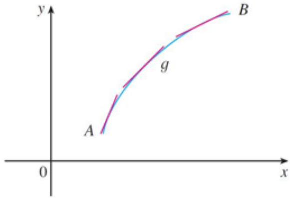
In Figure 6(a), the slope of the tangent increases, meaning that f' is an increasing function and therefore its derivative f'' is positive.
In Figure 6(b) the slope of the tangent decreases, so f' decreases and therefore f'' is negative.

Concavity Test
(a) If $f''(x) > 0$ for all x in I, then graph of f is concave upward on I.
(b) If $f''(x) < 0$ for all x in I, then graph of f is concave downward on I



Concave upward

Figure 6(a)



Concave downward

Figure 6(b)

The Second Derivative Test

$(x_0, f(x_0))$

Definition A point P on a curve $y = f(x)$ is called an **inflection point** if f is continuous at this point, and the curve changes from concave upward to concave downward or from concave downward to concave upward.

$x = c$

The Second Derivative Test Suppose f'' are continuous near c.
(a) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c
(b) If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c

$\exists h$, $x \in (c-h, c+h)$, $\Rightarrow f''(x) > 0$, $(c-h, c)$ $f'(x) \uparrow \Rightarrow f'(x) < 0$
$(c, c+h)$ $f'(x) \uparrow \Rightarrow f'(x) > 0$ $\Rightarrow$ local minimum @ $x = c$

$\exists h$, $x \in (c-h, c+h) \Rightarrow f''(x) < 0$, $(c-h, c)$ $f'(x) \downarrow \Rightarrow f'(x) > 0$
$(c, c+h)$ $f'(x) \downarrow \Rightarrow f'(x) < 0$ $\Rightarrow$ local maximum @ $x = c$

Understanding Checker

Q1. Point (0,0) is the inflection point for functions

(A) x^3, x^5 (B) x, x^3, (C) x^4, x^6, (D) x^2, x^3, x^4

*Check if point is a point in the function

*Check if concavity changes

$f(x)$	x	x^2	x^3	x^4	x^5	x^6
$f'(x)$	1	$2x$	$3x^2$	$4x^3$	$5x^4$	$6x^5$
$f''(x)$	0	2	$6x$	$12x^2$	$20x^3$	$30x^4$
$f''(x)$ when $x<0$:	constant		$-$	$+$	$-$	$+$
$f''(x)$ when $x>0$:	constant		$+$	$+$	$+$	$+$

Q2. Find the critical numbers for function $\frac{x^2}{x-3}$.

(A) 0, 3, 6 (B) 0, 6 (C) 3, 6 (D) 0, 3

Function is only well-defined in the domain $(-\infty, 3) \cup (3, \infty)$

$x=3$ can never be a critical number (critical number must be in the domain)

$\therefore$ (B)

$f(x) = \frac{x^2}{x-3}$

$f'(x) = \frac{2x(x-3)-x^2}{(x-3)^2} = \frac{x(x-6)}{(x-3)^2}$

Q3. For $g(x) = x|x|$, we have the following results. Which ones are TRUE?

i. $g''(0)$ exists

ii. $g''(0)$ D.N.E

iii. (0,0) is an inflection point

iv. (0,0) is not an inflection point

(A) i and iii (B) ii and iii (C) ii and iv (D) i and iv

$$g(x) = \begin{cases} x^2 & x>0 \\ 0 & x=0 \\ -x^2 & x<0 \end{cases} \qquad g'(x) = \begin{cases} 2x & x>0 \\ 0 & x=0 \\ -2x & x<0 \end{cases}$$

$$g'(0) \begin{cases} \lim_{x\to 0^+} \frac{x^2-0}{x-0} \\ \lim_{x\to 0^-} \frac{-x^2-0}{x-0} \end{cases} = 0$$

$$g''(x) = \begin{cases} 2 & x>0 \\ \text{D.NE} & x=0 \\ -2 & x<0 \end{cases} \qquad g''(0) = \begin{cases} \lim_{x\to 0^+} \frac{2x-0}{x-0} = 2 \\ \lim_{x\to 0^-} \frac{-2x-0}{x-0} = -2 \end{cases}$$

For $x>0$, $g''(x)>0$ and for $x<0$, $g''(x)<0$.

$\therefore$ There is a change in concavity. $\therefore$ It is an inflection point

The Second Derivative Test

Example: Discuss the concavity, points of inflection, and local maxima and minima of curve $y = x^4 - 4x^3$.

Solution: If $f(x) = x^4 - 4x^3$, then

$$f'(x) = 4x^3 - 12x^2 = 4x^2(x-3)$$

$$f''(x) = 12x^2 - 24x = 12x(x-2)$$

To find the critical numbers we set $f'(x) = 0$ and obtain $x = 0$ and $x = 3$. To use the Second Derivative Test we evaluate f at these critical numbers:

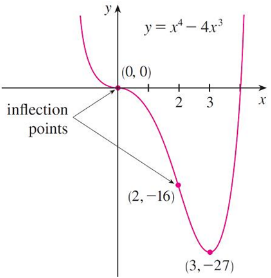
$$f''(0) = 0 \text{ and } f''(3) = 36 > 0$$

Since $f'(3) = 0$ and $f''(3) > 0$, $f(3) = -27$ is a local minimum.
(In fact, the expression for $f'(x)$ shows that $f(x)$ decreases to the left of 3 and increases to the right of 3.)

The Second Derivative Test

Since $f''(0) = 0$, the Second Derivative Test dose not work.
Since $f'(x) < 0$ for $x < 0$ and $0 < x < 3$, from First Derivative Test, $f(x)$ does not have a local maximum or minimum at 0.
Since $f''(x) = 0$ at $x = 0$ and 2, the real line can be divided into three intervals.
The point $(0, 0)$ and $(2, -16)$ are inflection points.

Interval	$f''(x) = 12x(x-2)$	Concavity
$(-\infty, 0)$	$+$	upward
$(0, 2)$	$-$	downward
$(2, \infty)$	$+$	upward



Note
The Second Derivative Test is inconclusive when $f''(c) = 0$. In other words, at such a point there might be a maximum, there might be a minimum, or there might be neither.

This test also fails when $f''(c)$ does not exist. In such cases the First Derivative Test must be used. In fact, even when both tests apply, the First Derivative Test is often the easier one to use.

The Second Derivative Test

Example: Sketch the graph of the function $f(x) = x^{2/3}(6-x)^{1/3}$.

Solution:
Calculation of the first two derivatives gives:

$$f'(x) = \frac{4-x}{x^{1/3}(6-x)^{2/3}}$$
$$f''(x) = \frac{-8}{x^{4/3}(6-x)^{5/3}}$$

Since $f'(x) = 0$ at $x = 4$, and $f'(x)$ does not exist at $x = 0$ and 6, the critical numbers are 0, 4 and 6.

Interval	$4-x$	$x^{1/3}$	$(6-x)^{2/3}$	$f'(x)$	f
$x < 0$	+	−	+	−	decreasing on $(-\infty, 0)$
$0 < x < 4$	+	+	+	+	increasing on $(0, 4)$
$4 < x < 6$	−	+	+	−	decreasing on $(4, 6)$
$x > 6$	−	+	+	−	decreasing on $(6, \infty)$

The Second Derivative Test

To find the local extreme values, we use the First Derivative Test.
Since $f'(x)$ changes from negative to positive at 0, $f(0) = 0$ is a local minimum.
Since $f'(x)$ changes from positive to negative at 4, $f(4) = 2^{5/3}$ is a local maximum.
The sign of $f'(x)$ does not change at 6, so there is no minimum or maximum there. (The Second Derivative Test could be used at 4, but not at 0 or 6, since f'' does not exist at either of these numbers.)

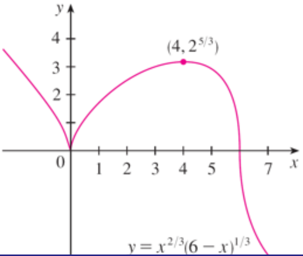
Interval	$4 - x$	$x^{1/3}$	$(6 - x)^{2/3}$	$f'(x)$	f
$x < 0$	+	−	+	−	decreasing on $(-\infty, 0)$
$0 < x < 4$	+	+	+	+	increasing on $(0, 4)$
$4 < x < 6$	−	+	+	−	decreasing on $(4, 6)$
$x > 6$	−	+	+	−	decreasing on $(6, \infty)$

The Second Derivative Test

For $f''(x) = \frac{-8}{x^{4/3}(6-x)^{5/3}}$, since $x^{4/3} \geq 0$ for all x, we have $f''(x) < 0$ for $x < 0$ and $0 < x < 6$, $f''(x) > 0$ for $x > 6$.

Interval	$4 - x$	$x^{1/3}$	$(6 - x)^{2/3}$	$f'(x)$	f
$x < 0$	+	−	+	−	decreasing on $(-\infty, 0)$
$0 < x < 4$	+	+	+	+	increasing on $(0, 4)$
$4 < x < 6$	−	+	+	−	decreasing on $(4, 6)$
$x > 6$	−	+	+	−	decreasing on $(6, \infty)$

Note that the curve has vertical tangents at $(0, 0)$ and $(6, 0)$ because $|f'(x)| \to \infty$ as $x \to 0$ and as $x \to 6$.



Let x become arbitrarily large (positive or negative) and see what happens to y. For a function f defined by

$$f(x) = \frac{x^2 - 1}{x^2 + 1}$$

The table gives values of this function, and the graph of f has been drawn by a computer in Figure 1.

x	$f(x)$
0	-1
± 1	0
± 2	0.600000
± 3	0.800000
± 4	0.882353
± 5	0.923077
± 10	0.980198
± 50	0.999200
± 100	0.999800
± 1000	0.999998

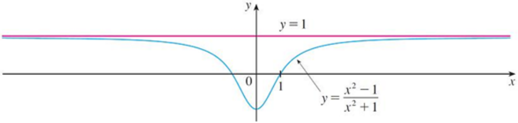


Figure 1

Limits at Infinity

As x grows larger and larger, you can see that the values of $f(x)$ get closer and closer to 1. In fact, it seems that we can make the values of $f(x)$ as close as we like to 1 by taking x sufficiently large.

Mathematically, this is

$$\lim_{x\to\infty} \frac{x^2-1}{x^2+1} = 1$$

Intuitive Definition of a Limit at Infinity

Let $f(x)$ be a function defined on some interval (a, ∞). Then

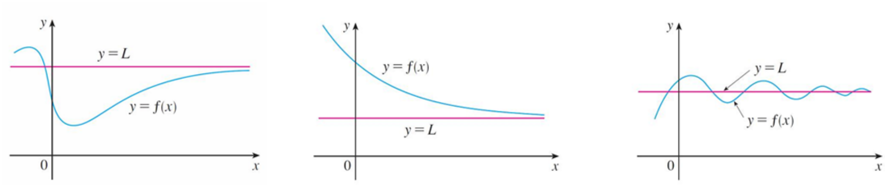
$$\lim_{x\to\infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by requiring x to be sufficiently large.

Another notation for $\lim_{x\to\infty} f(x) = L$ is

$$f(x) \to L \quad as \quad x \to \infty$$

Geometric illustrations of Definition 1 are shown in Figure 2.



Examples illustrating $\lim_{x\to\infty} f(x) = L$

Figure 2

Notice that there are many ways for the graph of f to approach the line $y = L$ (which is called a horizontal asymptote) as we look to the far right of each graph.

Limits at Infinity

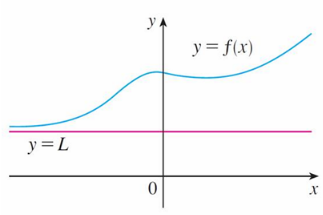
Definition 2

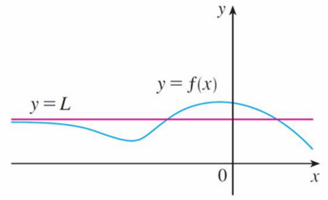
Let $f(x)$ be a function defined on some interval $(-\infty, a)$. Then

$$\lim_{x \to -\infty} f(x) = L$$

means that the value of $f(x)$ can be made arbitrarily close to L by requiring x to be sufficiently large negative.

Definition 2 is illustrated in the figure below. Notice that the graph approaches the line $y = L$ as $x \to -\infty$.





Definition 3

The line $y = L$ is called a **horizontal asymptote** of the curve $y = f(x)$ if either

only need one

$$\lim_{x\to\infty} f(x) = L \quad \text{or} \quad \lim_{x\to-\infty} f(x) = L$$

For $y = \frac{1}{x}$, we have

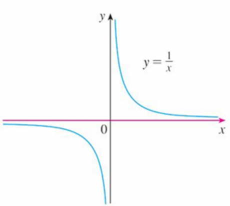
$$\lim_{x\to\infty} \frac{1}{x} = 0 \qquad \lim_{x\to-\infty} \frac{1}{x} = 0$$

Vertical Asymptote (V.A)

$\lim_{x\to a^+/a^-} f(x) = +\infty/-\infty$

$x=a$ is V.A

The line $y = 0$ (the x-axis) is a horizontal asymptote of the curve $y = \frac{1}{x}$.



Theorem 4

If $r > 0$ is a rational number, then

$$\lim_{x \to \infty} \frac{1}{x^r} = 0$$

If $r > 0$ is a rational number, such that x^r is defined for all x, then

$$\lim_{x \to -\infty} \frac{1}{x^r} = 0$$

$$\times \lim_{x \to -\infty} \frac{1}{\sqrt{x}}$$

Example: Evaluate $\lim_{x\to\infty} \frac{3x^2-x-2}{5x^2+4x+1}$.

Solution:
As x becomes large, both numerator and denominator become large, so it isn't obvious what happens to their ratio. We need to do some preliminary algebra.

To evaluate the limit at infinity of any rational function, we first divide both the numerator and denominator by the highest power of x that occurs in the denominator. (We may assume that $x \neq 0$ since we are interested only in large values of x.)

In this case the highest power of x in the denominator is x^2, so we have

$$\lim_{x\to\infty}\frac{3x^2-x-2}{5x^2+4x+1}=\lim_{x\to\infty}\frac{\frac{3x^2-x-2}{x^2}}{\frac{5x^2+4x+1}{x^2}}=\lim_{x\to\infty}\frac{3-\frac{1}{x}-\frac{2}{x^2}}{5+\frac{4}{x}+\frac{1}{x^2}}$$

$$=\frac{\lim_{x\to\infty}(3-\frac{1}{x}-\frac{2}{x^2})}{\lim_{x\to\infty}(5+\frac{4}{x}+\frac{1}{x^2})}$$

$$=\frac{\lim_{x\to\infty}3-\lim_{x\to\infty}\frac{1}{x}-2\lim_{x\to\infty}\frac{1}{x^2}}{\lim_{x\to\infty}5+4\lim_{x\to\infty}\frac{1}{x}+\lim_{x\to\infty}\frac{1}{x^2}}$$

$$=\frac{3-0-0}{5+0+0}$$

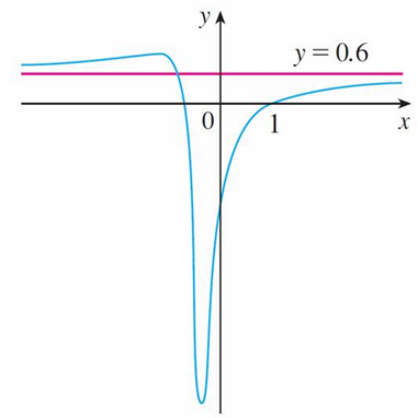
$$=\frac{3}{5}$$

$\lim_{x\to-\infty}$ still the same for this case

Limits at Infinity

A similar calculation shows that the limit as $x \to -\infty$ is also $\frac{3}{5}$.

Figure below illustrates the results of these calculations by showing how the graph of the given rational function approaches the horizontal asymptote $y = \frac{3}{5} = 0.6$.



Example: Find the horizontal and vertical asymptotes of the graph of the function

$$f(x) = \frac{\sqrt{2x^2+1}}{3x-5}$$

Solution: Dividing both numerator and denominator by x, we have

$$\lim_{x\to\infty} \frac{\sqrt{2x^2+1}}{3x-5} = \lim_{x\to\infty} \frac{\frac{\sqrt{2x^2+1}}{x}}{\frac{3x-5}{x}} = \lim_{x\to\infty} \frac{\sqrt{\frac{2x^2+1}{x^2}}}{\frac{3x-5}{x}}$$

$$= \frac{\lim_{x\to\infty} \sqrt{2+\frac{1}{x^2}}}{\lim_{x\to\infty}(3-\frac{5}{x})} = \frac{\sqrt{\lim_{x\to\infty} 2 + \lim_{x\to\infty} \frac{1}{x^2}}}{\lim_{x\to\infty} 3 - 5\lim_{x\to\infty} \frac{1}{x}} = \frac{\sqrt{2+0}}{3-5*0} = \frac{\sqrt{2}}{3}$$

Therefore the line $y = \frac{\sqrt{2}}{3}$ is a horizontal asymptote of the graph of f.

Horizontal Asymptotes

In computing the limit as $x \to -\infty$, we have $\sqrt{x^2} = |x| = -x$ $(x < 0)$. So when we divide the numerator $\sqrt{2x^2+1}$ by x, for $x < 0$ we get

$$\frac{\sqrt{2x^2+1}}{x} = \frac{\sqrt{2x^2+1}}{-\sqrt{x^2}} = -\sqrt{\frac{2x^2+1}{x^2}} = -\sqrt{2+\frac{1}{x^2}}$$

For the original equation, we have

$$\lim_{x\to-\infty} \frac{\sqrt{2x^2+1}}{3x-5} = \lim_{x\to-\infty} \frac{-\sqrt{2+\frac{1}{x^2}}}{3-\frac{5}{x}} = \frac{-\sqrt{2+\overbrace{\lim_{x\to-\infty}\frac{1}{x^2}}^{=0}}}{3-5\underbrace{\lim_{x\to-\infty}\frac{1}{x}}_{=0}} = -\frac{\sqrt{2}}{3}$$

Thus the line $y = -\frac{\sqrt{2}}{3}$ is also a horizontal asymptote.

Horizontal Asymptotes

A vertical asymptote is likely to occur when the denominator equals to 0, i.e., $3x - 5 = 0$. That is $x = \frac{5}{3}$.

The numerator $\sqrt{2x^2 + 1}$ is always nonnegative.
If x is close to $\frac{5}{3}$ and $x > \frac{5}{3}$, then $3x - 5$ is positive, thus $f(x)$ is positive.

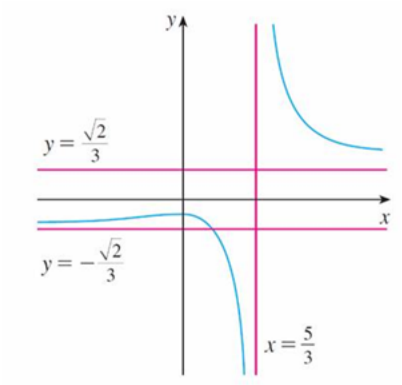
$$\lim_{x \to (\frac{5}{3})^+} \frac{\sqrt{2x^2+1}}{3x-5} = \infty$$

If x is close to $\frac{5}{3}$ and $x < \frac{5}{3}$, then $3x - 5$ is negative, thus $f(x)$ is negative.

$$\lim_{x \to (\frac{5}{3})^-} \frac{\sqrt{2x^2+1}}{3x-5} = -\infty$$

The vertical asymptote is $x = \frac{5}{3}$.

All three asymptotes are shown below,



$$y = \frac{\sqrt{2x^2+1}}{3x-5}$$

Figure 8

Infinite Limits at Infinity

The notation

$$\lim_{x \to \infty} f(x) = \infty$$

means $f(x)$ becomes large as x becomes large. Similarly we have,

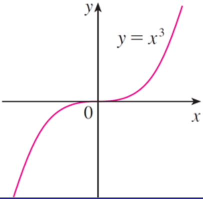
$$\lim_{x \to -\infty} f(x) = \infty \qquad \lim_{x \to \infty} f(x) = -\infty \qquad \lim_{x \to -\infty} f(x) = -\infty$$

Example: Find $\lim_{x \to \infty} x^3$ and $\lim_{x \to -\infty} x^3$.

Solution: From the graph, we have

$$\lim_{x \to \infty} x^3 = \infty \qquad \lim_{x \to -\infty} x^3 = -\infty$$

$\lim_{x \to \infty} \sin x \Rightarrow$ D.N.E, it oscillates



5 Precise Definition of a Limit at Infinity

Let $f(x)$ be a function defined on some interval (a, ∞). Then

treat as an Error

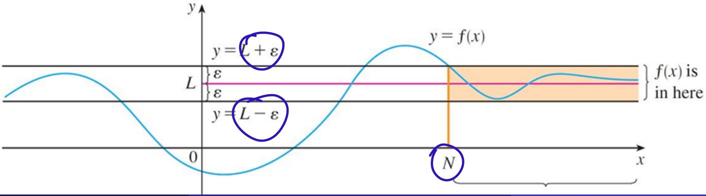
$$\lim_{x \to \infty} f(x) = L$$

$N(\varepsilon)$

means for every $\epsilon > 0$, there exists a corresponding number N, such that

$$\textit{if} \quad x > N \quad \textit{then} \quad |f(x) - L| < \epsilon$$

If x large enough, we can make the graph of $f(x)$ lie between the given horizontal lines $y = L - \epsilon$ and $y = L + \epsilon$.



Precise Definition of a Limit at Infinity

6 Definition

Let f be a function defined on some interval $(-\infty, a)$. Then

$$\lim_{x\to-\infty} f(x) = L$$

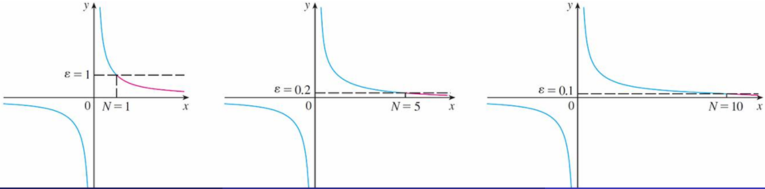
$\varepsilon, \delta \Rightarrow$ small values

$\rightarrow$ large value $\rightarrow M$ also

means for every $\epsilon > 0$, there is a corresponding number N, such that

$$\textit{if} \quad x < N \quad \textit{then} \quad \boxed{|f(x) - L| < \epsilon} \Rightarrow L - \varepsilon < f(x) < L + \varepsilon$$

For $\lim_{x\to\infty} \frac{1}{x} = 0$, some values of ϵ and the corresponding values of N can be given as below.



Example: Use Definition 5 to prove that $\lim_{x\to\infty} \frac{1}{x} = 0$.

Solution:
Given $\epsilon > 0$, we want to find N such that

$|f(x) - L|$

$$\textit{if} \quad x > N \quad \textit{then} \quad |\frac{1}{x} - 0| < \epsilon$$

In computing the limit we may assume that $x > 0$.
Then $\frac{1}{x} < \epsilon \quad \Longleftrightarrow \quad x > \frac{1}{\epsilon}$
Lets choose $N = \frac{1}{\epsilon}$. So $N(\varepsilon)$

$$\textit{if} \quad x > N = \frac{1}{\epsilon} \quad \textit{then} \quad |\frac{1}{x} - 0| = \frac{1}{x} < \epsilon$$

Therefore, by Definition,

$$\lim_{x\to\infty} \frac{1}{x} = 0$$

Precise Definition of an Infinite Limit at Infinity

7 Definition of an Infinite Limit at Infinity

Let f be a function defined on some interval (a, ∞). Then

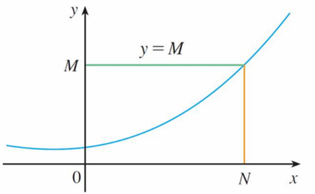
$$\lim_{x \to \infty} f(x) = \infty$$

means that for every positive number M, there is a corresponding number N, such that

$N(M)$

$$\textit{if} \quad x > N \quad \textit{then} \quad f(x) > M$$

Similar definitions apply when the symbol ∞ is replaced by $-\infty$.



Slant Asymptotes

Some asymptotes are *oblique*, that is, neither horizontal nor vertical.

must be a linear function.

The line $y = mx + b$ $(m \neq 0)$ is called a **slant asymptote**, if

Slant asymptotes can only happen when the numerator and denominator differs by 1 order. e.g. $\frac{x^3}{x^2}$, $\frac{x^5}{x^4}$ etc

$$\lim_{x\to\infty} [f(x) - (mx + b)] = 0$$

∞ $-\infty$ → just satisfy one of the conditions

If $m=0$

$\lim_{x\to\infty} [f(x) - b] = 0$

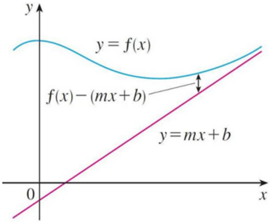
$\lim_{x\to\infty} f(x) = b$

$y = b$

this is horizontal asymptote

as the vertical distance between the curve $y = f(x)$ and the line $y = mx + b$ approaches 0.

For rational functions, slant asymptotes occur when the degree of the numerator is higher than that of the denominator by one.



Slant Asymptotes

The slant asymptote can be found by long division. For example, we have

$$f(x) = \frac{x^3}{x^2+1}.$$

From long division, we have

$$f(x) = \frac{x^3}{x^2+1} = x - \frac{x}{x^2+1}.$$

This equation suggests that $y = x$ is a candidate for a slant asymptote. In fact,

$\frac{1}{x} \to 0$ when $x \to \infty$

$$f(x) - x = -\frac{x}{x^2+1} = -\frac{\frac{1}{x}}{1+\frac{1}{x^2}} \to 0 \quad \textit{as } x \to \pm\infty$$

$\frac{1}{x^2} \to$ 0 when $x \to \infty$

$\to$ tells that $\frac{x}{x^2+1}$ does not play an impt role when $x \to \infty$

So the line $y = x$ is a slant asymptote.

Guidelines for Sketching a Curve

Checklist for sketching a curve $y = f(x)$ by hand.
Not every item is relevant. But the guidelines provide all the information you need to make a sketch of a function, that captures its most important properties.

A. Domain
It is often useful to start by determining the domain D of $f(x)$, on which $f(x)$ is defined.

B. Intercepts
$x=0$
The y-intercept is $f(0)$. To find the x-intercepts, we set $y = 0$ and solve for x. (Omit this step, if the function is difficult to solve.)

C. Symmetry
(i)*Even function* $f(-x) = f(x)$, for all x in D.
(ii)*Odd function* $f(-x) = -f(x)$, for all x in D.
(iii)*Periodic function* $f(x + p) = f(x)$, for all x in D.

D. Asymptotes

(i) **Horizontal Asymptotes**. If either $\lim_{x\to\infty} f(x) = L$ or $\lim_{x\to-\infty} f(x) = L$, then the line $y = L$ is a horizontal asymptote.

(ii) **Vertical Asymptotes**. The line $x = a$ is a vertical asymptote, if at least one of the following statements is true, $\lim_{x\to a^+/a^-/a} f(x) = +\infty/-\infty$

$$\lim_{x\to a^+} f(x) = \infty \qquad \lim_{x\to a^-} f(x) = \infty \qquad \lim_{x\to a^+} f(x) = -\infty \qquad \lim_{x\to a^-} f(x) = -\infty$$

If $f(a)$ is not defined, but a is an endpoint of the domain of $f(x)$, then you should compute $\lim_{x\to a^-} f(x)$ or $\lim_{x\to a^+} f(x)$, and check whether or not this limit is infinite.

(iii) **Slant Asymptotes**. The line $y = mx + b$ $(m \neq 0)$ is a slant asymptote if

$$\lim_{x\to\infty/-\infty} [f(x) - (mx + b)] = 0.$$

Guidelines for Sketching a Curve

E. Intervals of Increase or Decrease
Use the I/D Test. Compute $f'(x)$ and find the intervals on which $f'(x)$ is positive (f is increasing) and the intervals on which $f'(x)$ is negative (f is decreasing).

F. Local Maximum and Minimum Values
Find the critical numbers of $f(x)$ (the numbers c where $f'(c) = 0$ or $f'(c)$ does not exist).

Then use the First Derivative Test. If f' changes from positive to negative at a critical number c, then $f(c)$ is a local maximum. If f' changes from negative to positive at c, then $f(c)$ is a local minimum.

Although it is usually preferable to use the First Derivative Test, you can use the Second Derivative Test. If $f'(c) = 0$ and $f''(c) \neq 0$, then $f''(c) > 0$ implies that $f(c)$ is a local minimum, whereas $f''(c) < 0$ implies that $f(c)$ is a local maximum.

Guidelines for Sketching a Curve

G. Concavity and Points of Inflection
Compute $f''(x)$ and use the Concavity Test. The curve is concave upward where $f''(x) > 0$, and concave downward where $f''(x) < 0$. Inflection points occur where the direction of concavity changes.

H. Sketch the Curve
Using the information in items **A - G** to draw the graph.

Sketch the asymptotes in **D** as dashed lines.

Plot the intercepts, maximum and minimum points, and inflection points.

Make the curve pass through these points, rising and falling according to **E**, with concavity according to **G**, and approaching the asymptotes in **D**.

If additional accuracy is required near any point, you can compute the value of its derivative. The tangent indicates the direction in which the curve proceeds.

Example: Use the guidelines to sketch the curve $y = \frac{2x^2}{x^2-1}$.
Solution: **A**. The domain is

$$\{x|\ x^2 - 1 \neq 0\} \to \{x|\ x \neq \pm 1\} \to (-\infty, -1) \cup (-1, 1) \cup (1, \infty)$$

B. The x- and y-intercepts are both 0. $y(0) = 0$

C. Function $f(x)$ is even $(f(-x) = f(x))$. The curve is symmetric.

$$f(-x) = \frac{2(-x)^2}{(-x)^2 - 1} = \frac{2x^2}{x^2-1} = f(x)$$

D.

$$\lim_{x\to\pm\infty} \frac{2x^2}{x^2-1} = \lim_{x\to\pm\infty} \frac{2}{1 - \underbrace{\frac{1}{x^2}}_{\to 0 \text{ when } x\to\pm\infty}} = 2.$$

Therefore the line $y = 2$ is a horizontal asymptote.

*since numerator and denominator have same order, there is no slant asymptote

$$\lim_{x\to 1^+} \frac{2x^2 \ {>0}}{x^2 - 1 \ {>0}} = \infty \qquad \lim_{x\to 1^-} \frac{2x^2 \ {>0}}{x^2 - 1 \ {<0}} = -\infty$$

$$\lim_{x\to -1^+} \frac{2x^2 \ {>0}}{x^2 - 1 \ {<0}} = -\infty \qquad \lim_{x\to -1^-} \frac{2x^2 \ {>0}}{x^2 - 1 \ {>0}} = \infty$$

Therefore the lines $x = -1$ and $x = 1$ are vertical asymptotes.

Guidelines for Sketching a Curve

↗ ↗ ↘ ↘

-1 0 1

E. $f'(x) = \frac{(x^2-1)(4x)-2x^2\cdot 2x}{(x^2-1)^2} = \frac{-4x}{(x^2-1)^2}$. Since $f'(x) > 0$ when $x < 0$ $(x \neq -1)$, and $f'(x) < 0$ when $x > 0$ $(x \neq 1)$, $f(x)$ is increasing on $(-\infty, -1)$ and $(-1, 0)$, and decreasing on (0, 1) and $(1, \infty)$.

→ critical number

F. The only critical number is $x = 0$ (1 and -1 are not in the domain of $f(x)$). Since f' changes from positive to negative at 0, $f(0) = 0$ is a local maximum by the First Derivative Test.

G. $f''(x) = \frac{(x^2-1)^2(-4)+4x\cdot 2(x^2-1)2x}{(x^2-1)^4} = \frac{12x^2+4}{(x^2-1)^3}$. Since $12x^2 + 4 > 0$ for all x, we have

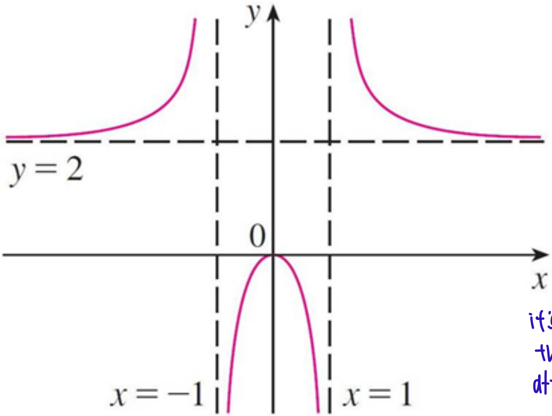
CU CD concave down concave up

-1 0 1

$$f''(x) > 0 \iff x^2 - 1 > 0 \iff |x| > 1$$

and $f''(x) < 0 \iff |x| < 1$. Thus the curve is concave upward on the intervals $(-\infty, -1)$ and $(1, \infty)$, and concave downward on $(-1, 1)$. It has no point of inflection, since 1 and -1 are not in the domain of $f(x)$.

H. Using the information in **A - G**, we can finish the sketch.



it's symmetrical about the y-axis, just flip it after we get the left side.

Finished sketch of $y = \dfrac{2x^2}{x^2 - 1}$

Guidelines for Sketching a Curve

Example: Sketch the graph of $f(x) = \frac{x^3}{x^2+1}$.

Solution:

A. The domain is $\mathbb{R} = (-\infty, \infty)$.

B. The x- and y-intercepts are both 0.

C. Since $f(-x) = -f(x)$, $f(x)$ is an odd function, and its graph is symmetric about the origin.

D. Since $x^2 + 1$ is never 0, there is no vertical asymptote. Since $f(x) \to \infty$ as $x \to \infty$ and $f(x) \to -\infty$ as $x \to -\infty$, there is no horizontal asymptote. The line $y = x$ is a slant asymptote.

E. $f'(x) = \frac{3x^2(x^2+1)-x^3\cdot 2x}{(x^2+1)^2} = \frac{x^2(x^2+3)}{(x^2+1)^2}$

Since $f'(x) > 0$ for all x (except 0), f is increasing on $(-\infty, \infty)$.

F. Although $f'(0) = 0$, f' does not change sign at 0, so there is no local maximum or minimum.

G. $f''(x) = \frac{(4x^3+6x)(x^2+1)^2-(x^4+3x^2)\cdot 2(x^2+1)2x}{(x^2+1)^4} = \frac{2x(3-x^2)}{(x^2+1)^3}$

Since $f''(x) = 0$ when $x = 0$ or $x = \pm\sqrt{3}$, we set up the following chart:

Interval	x	$3 - x^2$	$(x^2 + 1)^3$	$f''(x)$	f
$x < -\sqrt{3}$	$-$	$-$	$+$	$+$	CU on $(-\infty, -\sqrt{3})$
$-\sqrt{3} < x < 0$	$-$	$+$	$+$	$-$	CD on $(-\sqrt{3}, 0)$
$0 < x < \sqrt{3}$	$+$	$+$	$+$	$+$	CU on $(0, \sqrt{3})$
$x > \sqrt{3}$	$+$	$-$	$+$	$-$	CD on $(\sqrt{3}, \infty)$

The points of inflection are $(-\sqrt{3}, -\frac{3}{4}\sqrt{3})$, $(0, 0)$, and $(\sqrt{3}, \frac{3}{4}\sqrt{3})$

H. The graph of f is sketched in Figure 11.

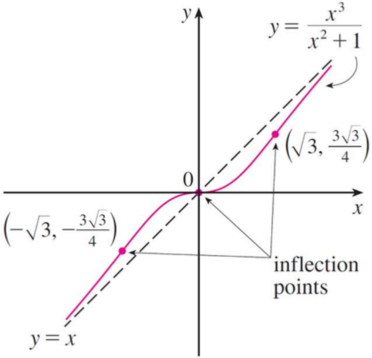


Figure 11